



Urban Traffic Accident Risk Assessment and Prediction Using Machine Learning

Anjali Kasarla • Vinusha Muppala • Lasya Geethika

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Under the guidance of Dr. Katanguri Swanthana (CSE)

ABSTRACT

Urban traffic accidents remain a major public-safety challenge because they produce fatalities, injuries, congestion, and economic losses. Accident-related information is generated in both structured and unstructured forms, including accident reports, road and traffic attributes, weather information, news articles, and traffic-related social-media text.

Conventional traffic-management systems, however, commonly emphasize structured historical information and provide limited support for extracting predictive signals from free-form text. This paper presents a design-oriented machine-learning framework for urban traffic accident-risk prediction using text analysis.

The framework integrates Natural Language Processing (NLP), classical supervised learning, contextual language representation, and Explainable Artificial Intelligence (XAI) to classify traffic situations into Low, Medium, and High Risk.

The proposed pipeline performs text normalization, tokenization, stop-word removal, punctuation handling, stemming and lemmatization for classical NLP branches, followed by feature representation using Bag-of-Words, TF-IDF, Word2Vec/GloVe-style embeddings, and BERT. Candidate classifiers include Logistic Regression, Support Vector Machine, Naïve Bayes, Decision Tree, Random Forest, and BERT-based classification.

Model evaluation is defined using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix, with class imbalance addressed where necessary. SHAP and LIME are incorporated to identify influential features and textual evidence behind predictions. The final design targets deployment through Flask or FastAPI with a web dashboard.

Keywords: traffic accident risk prediction; natural language processing; TF-IDF; BERT; machine learning; explainable AI; SHAP; LIME; urban traffic safety.

1. INTRODUCTION

1.1 Background

Urban transportation systems generate heterogeneous information at high volume and frequency. Road geometry, traffic conditions, weather, time, vehicle characteristics, and human factors influence the probability and consequences of crashes. At the same time, textual incident descriptions contain contextual evidence that may not be represented explicitly in a traditional tabular schema. The project supplied for this study therefore frames accident-risk prediction as a text-analysis problem embedded within a broader traffic-safety pipeline.

Machine learning has become an established approach for crash occurrence, crash frequency, and injury-severity modeling. Recent reviews emphasize both the potential of ML and persistent methodological challenges, including class imbalance, evaluation design, non-crash sampling, data heterogeneity, temporal generalization, and real-time deployment. [1], [2] These issues motivate a design that is explicit about preprocessing, target definition, leakage prevention, model comparison, and explanation.

1.2 Problem Statement

The research problem is to develop a practical and explainable framework that converts accident-related text and relevant contextual variables into a three-class urban traffic accident-risk prediction. The desired output is a risk level rather than a simple binary accident/non-accident decision or a post-crash injury-severity estimate. The system should also provide human-understandable evidence for its decision.

- Unstructured accident-related text is underused compared with structured traffic and crash variables.
- Existing crash-prediction studies often optimize a single modeling family or a single data modality.
- Risk prediction and accident severity are related but distinct targets and should not be conflated.
- High-performing models may be difficult for practitioners to trust without local explanations.
- Offline research prototypes are not always translated into usable decision-support interfaces.

1.3 Research Gap

The supplied project presentation identifies five gaps: fragmented use of traffic, weather, road, and accident factors; limited use of textual accident information; limited explainability; limited integrated Low/Medium/High risk classification; and limited real-time decision support. These gaps are consistent with broader review literature emphasizing the need for better data integration, stronger evaluation, and practical deployment. [1], [2], [3]

1.4 Research Objectives

1. Develop an integrated machine-learning framework for urban traffic accident-risk prediction.
2. Use NLP to transform accident-related text into predictive representations.
3. Compare classical ML models with a BERT-based deep-learning approach.
4. Combine textual and contextual evidence where the final dataset schema permits.
5. Introduce leakage-controlled feature selection for high-dimensional sparse representations.
6. Classify observations into Low, Medium, and High Risk.
7. Evaluate models using multiple complementary performance measures.
8. Generate local explanations using SHAP and LIME.
9. Expose the selected model through a web-based prediction interface.

1.5 Research Questions

10. RQ1. Can accident-related text provide useful predictive information for urban traffic accident-risk classification?
11. RQ2. How do TF-IDF, traditional word embeddings, and contextual BERT representations compare for the task?
12. RQ3. Which candidate classifier provides the best balance of predictive performance, computational cost, and interpretability?
13. RQ4. Does feature selection improve generalization and reduce unnecessary sparse-text dimensions?
14. RQ5. Which textual and contextual factors most strongly influence the predicted risk class?
15. RQ6. Can the resulting pipeline be operationalized as an explainable decision-support dashboard?

1.6 Contributions

- A unified NLP-plus-context framework for three-level accident-risk classification.
- A reproducible comparison between classical sparse-text models and BERT-based classification.
- A leakage-aware feature-selection protocol using training data only.
- An explainability layer based on SHAP and LIME.
- A deployment-oriented architecture using Flask/FastAPI and a web dashboard.
- A transparent distinction between what is supported by the supplied project materials and what must be validated experimentally.

1.7 Organization of the Paper

Section 2 reviews existing approaches, machine-learning and deep-learning methods, explainability techniques, and the research gap. Section 3 describes the proposed methodology and mathematical formulation. Section 4 defines the experimental setup. Section 5 provides the results-reporting framework and explicitly identifies which empirical outputs remain pending. Section 6 discusses how findings should be interpreted once experiments are completed. Sections 7 and 8 present limitations and conclusions/future work, followed by declarations, acknowledgements, and references.

2. RELATED WORK

2.1 Existing Approaches

Road-safety prediction has historically used statistical models, machine learning, deep learning, and sensor-based detection. Recent systematic reviews classify the literature into crash occurrence or real-time prediction, crash frequency prediction, and injury-severity prediction. [1], [2] These categories differ in target definition, data requirements, and evaluation protocol.

Approach	Typical inputs	Strength	Limitation for this project
Historical crash modeling	Crash records, road attributes	Established baseline and interpretable factors	May not exploit unstructured text
Real-time detector models	Speed, volume, occupancy, detector streams	Supports temporal risk estimation	Infrastructure dependent
Spatiotemporal DL	Crash mobility + weather + spatial data	Captures spatial/temporal dependencies	Higher complexity; transfer can be difficult
Severity prediction	Post-crash attributes	Useful for emergency planning	Not equivalent to proactive risk
CCTV / vision detection	Video streams	Direct incident detection	Requires camera infrastructure; often post-event

2.2 Machine Learning Approaches

Classical machine learning remains important because sparse text representations such as TF-IDF can contain tens of thousands of dimensions while remaining computationally manageable for linear models. Logistic Regression provides a strong probabilistic baseline; SVM is well suited to high-dimensional classification; Naïve Bayes is a classic text-classification baseline; and tree ensembles such as Random Forest can capture nonlinear relationships among engineered and contextual variables. [4]-[8]

The literature review supplied with the project specifically includes Logistic Regression, SVM, Naïve Bayes, Decision Tree, and Random Forest. Random Forest has also shown strong performance in reviews of crash-severity modeling, although severity performance should not automatically be interpreted as evidence of future-risk superiority. [9]

2.3 Deep Learning Approaches

Deep learning has expanded crash prediction through temporal and spatial architectures. Bao et al. proposed a spatiotemporal deep-learning architecture using crash, taxi GPS, road, land-use, population, and weather data and reported that the approach could capture spatial and temporal characteristics in citywide short-term crash-risk prediction. [10] Yuan et al. used an LSTM-RNN for real-time crash-risk prediction and applied SMOTE to address

class imbalance. [11] LSTM provides a mechanism for learning dependencies over sequences, while Transformer-based models offer a different approach based on attention.

BERT is particularly relevant to this project because accident descriptions are contextual language. BERT uses bidirectional Transformer representations and can be fine-tuned for classification with a task-specific output layer. [12], [13] The trade-off is computational cost and the need for careful preprocessing, fine-tuning, and validation.

2.4 Explainability / Existing Techniques

Explainability is essential when a risk prediction may influence operational decisions. LIME explains an individual prediction by fitting an interpretable local approximation around the instance. [14] SHAP provides additive feature-attribution values based on Shapley-value principles and can be used to quantify how inputs contribute to a particular prediction. [15] In this project, the explanations should be treated as model-attribution evidence rather than causal proof.

2.5 Critical Comparison

Study	Model method	Data setting	Relevance	Limitation / gap
Bao et al. (2019) [10]	STCL-Net / CNN / LSTM	NYC multi-source	Multi-source spatiotemporal risk prediction	Geographic and data-pipeline dependence
Yuan et al. (2019) [11]	LSTM-RNN + SMOTE	Traffic detector + crash	Real-time temporal modeling	Infrastructure dependence; limited text/XAI
Nikolaou et al. (2023) [16]	LR, DT, RF, SVM, KNN	Motorway segments	Risk-level classification	Motorway-specific setting
Yan & Shen (2022) [17]	Random Forest + optimization	US-Accidents	Large accident dataset	Severity target rather than future risk
Ali et al. (2024) [1]	Systematic review	Crash ML literature	Identifies methodological gaps	Calls for stronger data/evaluation practice
Alhawarat et al. (2026) [18]	Gradient boosting + SHAP + calibration	US + UK crash datasets	Explainability and transfer evaluation	Severity target; transfer still challenging

2.6 Research Gap

The proposed research occupies the intersection of three strands that are often studied separately: textual incident analysis, proactive risk-level classification, and explainable deployment. The novelty claim is therefore intentionally scoped: the project proposes an integrated, reproducible pipeline rather than claiming a new neural architecture. The empirical contribution will be the comparison of representations and models under a common target definition and evaluation protocol.

3. PROPOSED METHODOLOGY

3.1 Overall Framework

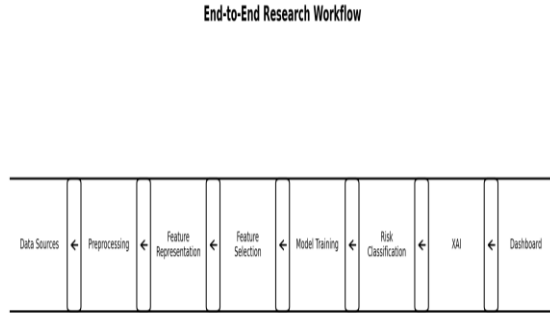


Figure 1. End-to-end research workflow.

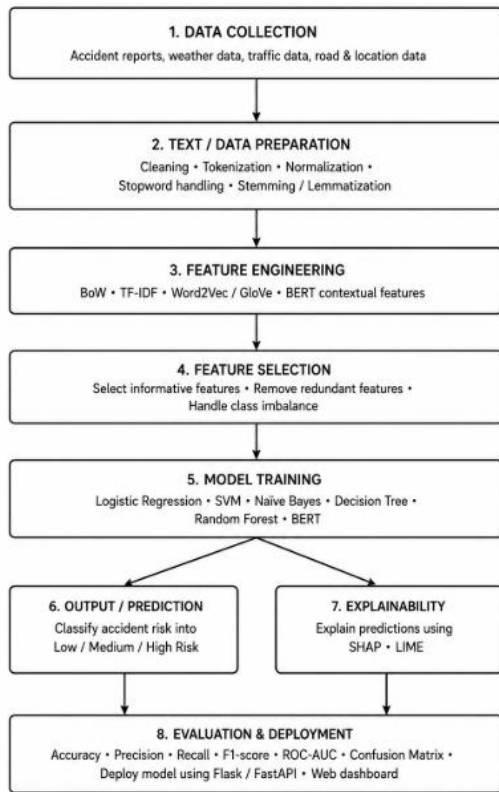


Figure 2. Proposed system architecture.

3.2 Dataset

The supplied abstract identifies the Kaggle Road Accident dataset and a Kaggle search for traffic-accident Twitter data. The Road Accident dataset description indicates fields covering injury type, vehicle type, time of day, road type, geographical area, and road conditions. [19] The project abstract additionally proposes accident reports, traffic tweets, news, weather, road, and traffic information as possible sources.

Dataset / source	Intended role	Known status
Kaggle Road Accident Dataset [19]	Primary structured	Explicitly supplied in project abstract

Dataset / source	Intended role	Known status
	accident/context dataset	
Traffic-accident social-media text	Unstructured incident language	Referenced in supplied abstract/PPT
Accident reports / incident descriptions	Primary NLP input	Proposed source type
Weather / road / traffic fields	Context variables	Specified conceptually in project materials

Because the supplied materials do not provide a final merged schema, exact sample count, missingness profile, or authoritative Low/Medium/High labeling rule, these values must be established during implementation. A professional manuscript should not silently invent them.

3.3 Data Preprocessing

Stage	Operation	Rationale
Text normalization	Lowercasing, whitespace normalization	Reduce superficial variation
Tokenization	Word/subword tokenization	Convert text into model inputs
Noise handling	Punctuation / URL / irrelevant-symbol handling	Reduce non-semantic noise where justified
Stop words	Remove selectively for classical NLP	Reduce common low-information tokens
Stemming / lemmatization	Classical branch only	Normalize lexical variants
Missing values	Imputation or explicit missing category	Avoid invalid inputs
Duplicates	Deduplication / near-duplicate review	Reduce leakage and repeated evidence
Split	Train/validation/test before learned transformations	Prevent leakage

BERT should use its native tokenizer rather than applying aggressive stemming or lemmatization. Similarly, resampling methods such as SMOTE, if needed, must be applied only to the training partition. These decisions are important because preprocessing can otherwise leak information or destroy contextual language.

3.4 Feature Engineering

Four primary representation families are planned: BoW, TF-IDF, word embeddings, and BERT. TF-IDF is the principal classical baseline because it converts a document into a sparse weighted vector that emphasizes discriminative terms. [20] Word2Vec-style embeddings represent words in a continuous vector space, while GloVe provides an alternative global-statistics embedding. [21], [22]

Feature family	Representation	Example use
BoW	Term counts	Baseline text classification
TF-IDF	Weighted sparse vector	Primary classical text baseline

Feature family	Representation	Example use
Word embeddings	Dense semantic vectors	Semantic feature comparison
BERT	Contextual token/document representation	Deep-learning comparison
Context features	Weather, road, traffic, time	Complement textual evidence
Text statistics	Length, token count, domain-term counts	Low-dimensional engineered features

3.5 Feature Selection

For sparse TF-IDF features, the proposed selection method is SelectKBest using mutual information. The selector is fitted only on training data. The value of K should be tuned using validation or cross-validation. This is a methodological recommendation, not a fact specified in the original project abstract.

```
from sklearn.feature_selection import SelectKBest,
mutual_info_classif
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression

selector = SelectKBest(
    score_func=mutual_info_classif,
    k=3000
)

pipeline = Pipeline([
    ("select", selector),
    ("clf", LogisticRegression(
        max_iter=2000,
        class_weight="balanced"
    ))
])

pipeline.fit(X_train_tfidf, y_train)
y_pred = pipeline.predict(X_test_tfidf)
```

The selection stage should be evaluated as an ablation: compare the same classifier with all candidate features against the selected subset. Feature selection should improve efficiency and/or generalization; it should not be assumed to improve accuracy.

3.6 Proposed Model

The proposed primary model is a comparative ensemble of candidate classifiers rather than a single predetermined winner. The final model is selected after validation using predictive performance, class-wise behavior, computational cost, and explainability. For the text branch, the recommended sequence is TF-IDF + Logistic Regression, TF-IDF + SVM, Naïve Bayes, followed by a BERT classifier. For mixed engineered/context features, Decision Tree and Random Forest provide nonlinear baselines.

3.7 Baseline Models

Baseline	Expected purpose
Majority-class baseline	Quantify the difficulty of the task and expose class imbalance
Logistic Regression	Linear sparse-text baseline
Linear SVM	Strong high-dimensional text baseline
Multinomial Naïve Bayes	Traditional probabilistic text classifier
Decision Tree	Interpretable nonlinear baseline
Random Forest	Robust nonlinear ensemble
BERT classifier	Contextual deep-learning comparison

3.8 Explainability Method

For tree or tabular models, SHAP can quantify feature contributions at the instance level. For text classifiers, explanation output should be presented carefully because token-level attributions may depend on the explainer and model architecture. LIME can provide a local approximation by perturbing the input and fitting an interpretable surrogate. [14], [15] The dashboard should show the predicted class, model confidence where meaningful, and a concise ranked list of influential evidence.

3.9 Algorithm / Mathematical Formulation

Let each observation be represented by x_i and its target risk class by $y_i \in \{\text{Low, Medium, High}\}$. For a TF-IDF representation, the weight of term t in document d can be expressed as:

$$tfidf(t,d) = tf(t,d) \times \log(N / (df(t) + 1))$$

where $tf(t,d)$ is the term frequency, $df(t)$ is the number of documents containing term t , and N is the number of documents. For multiclass Logistic Regression, the class probability can be represented using the softmax function:

$$P(y=k|x) = \exp(w_k^T x + b_k) / \sum_j \exp(w_j^T x + b_j)$$

The training objective minimizes regularized cross-entropy. For a class-balanced implementation, class weights can be used when the observed class distribution is substantially skewed. If resampling is used, it must be restricted to training data. SMOTE is one established approach for generating synthetic minority examples, but its appropriateness for sparse text must be empirically verified rather than assumed. [23]

4. EXPERIMENTAL SETUP

4.1 Hardware and Software

The implementation can be executed in Python using pandas/numpy for data handling, scikit-learn for classical ML, and Transformers/PyTorch or an equivalent framework for BERT. Scikit-learn provides a consistent API for preprocessing, model training, feature selection, and evaluation. [24] Exact hardware should be reported after execution.

Component	Recommended reporting
Programming language	Python version

Component	Recommended reporting
Classical ML	scikit-learn version
Deep learning	PyTorch/Transformers versions
XAI	SHAP and LIME versions
Compute	CPU model, RAM, GPU model/VRAM if used
Deployment	Flask or FastAPI version; browser/client details if relevant

4.2 Dataset Split

A stratified train/validation/test split is recommended when observations are independent. If timestamps are available and the research objective is forecasting, a temporal split is preferable to a random split because it better reflects deployment. The final manuscript should report the exact split ratio and random seed. No ratio is claimed here because it was not specified in the supplied project materials.

4.3 Hyperparameter Configuration

Model	Parameters to tune
Logistic Regression	C, regularization, class_weight
SVM	C, kernel/linear setting, class_weight
Naïve Bayes	alpha
Decision Tree	max_depth, min_samples_split, criterion
Random Forest	n_estimators, max_depth, max_features, min_samples_leaf
TF-IDF	ngram_range, min_df, max_df, sublinear_tf
Feature selection	K
BERT	learning rate, batch size, epochs, max sequence length, warmup, weight decay

4.4 Cross-Validation

Five-fold stratified cross-validation is recommended for the classical models when the dataset structure permits it. Hyperparameter selection should be nested within the training portion so that the final test set remains untouched. If temporal dependence exists, time-aware validation should replace random stratification.

4.5 Evaluation Metrics

The project specifies Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Because the task is multiclass, macro-averaged metrics should be reported alongside weighted averages. Macro-F1 gives each risk class equal importance and therefore helps expose poor performance on a minority class.

Metric	Definition / purpose
Accuracy	Overall proportion of correct predictions
Precision	Reliability of predictions assigned to a class
Recall	Ability to recover instances belonging to a class
F1-score	Harmonic mean of precision and recall
ROC-AUC	Ranking/discrimination measure, reported using an explicit multiclass strategy
Confusion Matrix	Class-by-class error structure
PR analysis	Useful when classes are imbalanced and positive-class performance matters

4.6 Statistical Testing

Model comparisons should not rely solely on a single test-set number. Repeated cross-validation scores can be compared using paired statistical procedures appropriate to the experimental design. For two classifiers evaluated on paired folds, a paired test may be used; for multiple models, a repeated-measures approach or Friedman test followed by an appropriate post-hoc test can be considered. The final choice should be reported with assumptions and effect sizes. The statistical test itself should not be selected after inspecting which model wins.

5. RESULTS

Important reporting note: the supplied abstract and presentation contain no executed model outputs, metric values, confusion matrices, ROC/PR curves, feature-importance plots, or statistical-test results. Accordingly, this section is structured as a publication-ready results framework but does not invent numerical findings. Once the implementation is run, the placeholders below should be replaced with measured values and figures.

5.1 Overall Performance

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	ROC-AUC
Majority baseline	TBD	TBD	TBD	TBD	TBD
Logistic Regression	TBD	TBD	TBD	TBD	TBD
SVM	TBD	TBD	TBD	TBD	TBD
Naïve Bayes	TBD	TBD	TBD	TBD	TBD
Decision Tree	TBD	TBD	TBD	TBD	TBD
Random Forest	TBD	TBD	TBD	TBD	TBD
BERT	TBD	TBD	TBD	TBD	TBD

5.2 Model Comparison

The completed comparison should identify the model that provides the strongest overall balance rather than simply the highest accuracy. Particular attention should be given to High Risk recall because a false negative in this class may be operationally more consequential than a false positive. This criterion should be defined before examining results.

5.3 Confusion Matrix

The confusion matrix should be reported for every leading model, with special emphasis on Low→Medium, Medium→High, and High→Low errors. The High→Low cell is especially important because it represents a potentially severe underestimation of risk.

5.4 ROC/PR Analysis

Multiclass one-vs-rest ROC curves and macro/weighted AUC values should be reported where probability outputs are reliable. Precision–Recall curves should also be considered when class imbalance is substantial. ROC interpretation should follow established guidance rather

than treating AUC as a complete substitute for class-specific error analysis. [25], [26]

5.5 Feature Importance

For classical models, the paper should report either model coefficients, permutation importance, or SHAP values. For tree models, both global importance and local explanations are recommended. For text models, the most influential words/phrases should be reported with examples, while avoiding the claim that an attribution proves causality.

5.6 Explainability Results

The completed study should include at least three representative examples: a Low Risk prediction, a Medium Risk prediction, and a High Risk prediction. Each example should show the input text, predicted class, confidence/probability if available, and the top positive/negative contributing features.

5.7 Ablation Study

Experiment	Change	Purpose	Result
A1	TF-IDF without feature selection	Baseline dimensionality	TBD
A2	TF-IDF + SelectKBest	Measure selection effect	TBD
A3	Text only	Quantify textual contribution	TBD
A4	Context only	Quantify structured/context contribution	TBD
A5	Text + context	Measure integration benefit	TBD
A6	Best classical model vs BERT	Measure representation/model gain	TBD
A7	Without XAI layer	Measure deployment overhead	TBD

5.8 Statistical Results

After repeated validation, report the mean, standard deviation, confidence interval where appropriate, and the selected significance test. If a model difference is statistically significant but practically small, the discussion should say so. No statistical conclusion is made here because the experiment has not yet been executed.

6. DISCUSSION

6.1 Interpretation of Findings

Once results are available, interpretation should connect model behavior to the research questions. A strong model with high macro-F1 and strong High Risk recall would support the feasibility of the proposed classification task. If BERT substantially outperforms sparse-text baselines, this would suggest that contextual language representation is valuable. If the difference is small, the simpler model may be preferable because of lower computational and deployment cost.

6.2 Comparison With Previous Studies

The results should be compared by task and data regime rather than by raw accuracy alone. Bao et al. studied citywide short-term risk with spatial and temporal multi-source data, while Yuan et al. focused on real-time sequence prediction using traffic detectors. [10], [11] Nikolaou et al. explicitly examined risk-level classification and compared several classical ML models. [16] Yan and Shen focused on severity rather than future risk. [17] These distinctions should be preserved when positioning the present work.

6.3 Theoretical Implications

The principal methodological implication is that accident-risk prediction can be framed as a multimodal evidence-integration problem in which unstructured text is treated as a predictive source rather than merely descriptive metadata. The framework also emphasizes that model evaluation, target definition, and explainability are part of the modeling contribution rather than post-processing additions.

6.4 Practical Implications

A deployed system could support traffic-management personnel by flagging potentially high-risk descriptions or contexts for additional attention. It should be used as decision support rather than an autonomous safety authority. The dashboard should communicate uncertainty, data freshness, and explanation evidence to avoid overconfidence.

6.5 Unexpected Findings

This subsection should be populated only after experimentation. Appropriate unexpected findings might include a classical TF-IDF model outperforming BERT, feature selection reducing performance, or a model showing high overall accuracy but poor High Risk recall. Such outcomes should be treated as scientifically useful rather than as failures, provided they are validated and explained.

7. LIMITATIONS

- The final risk-label construction is not specified in the supplied materials and must be documented before training.
- The supplied project files do not provide a finalized merged dataset, sample count, missing-value profile, or train/test statistics.
- Text from social media or news can contain noise, duplicates, sarcasm, location ambiguity, and reporting bias.
- Geographic transfer may be limited because traffic behavior, infrastructure, weather, and reporting practices vary by region.
- Class imbalance may make accuracy misleading; macro-F1 and class-specific recall are therefore important.

- SHAP and LIME are model-explanation methods, not causal inference methods.
- BERT may increase computational cost and latency relative to classical sparse-text models.
- A web dashboard does not by itself establish real-time validity; live data ingestion and freshness must be engineered separately.
- The present manuscript cannot report empirical performance until the proposed experiments are executed.

8. CONCLUSION AND FUTURE WORK

This paper presents a professional design for a machine-learning framework that predicts urban traffic accident risk from accident-related text and contextual information. The framework integrates NLP preprocessing, TF-IDF/BoW and embedding-based representations, BERT, classical machine-learning classifiers, feature selection, SHAP/LIME explanations, rigorous evaluation, and web deployment. The design responds directly to the gaps identified in the supplied project materials: fragmented factor analysis, limited use of text, limited explainability, lack of integrated risk levels, and limited decision-support deployment.

The principal next step is empirical execution. The team should finalize the dataset schema and risk-labeling rule, implement leakage-controlled preprocessing, train the baseline and advanced models, evaluate class-wise performance, perform the ablation study, generate SHAP/LIME explanations, and validate the deployment pipeline. Future research can extend the framework with temporal forecasting, geospatial modeling, multilingual traffic text, calibrated probabilities, cross-city validation, streaming data, and stronger explanation-faithfulness tests. Recent research also highlights the importance of calibration and cross-jurisdiction evaluation in practical crash prediction systems. [18]

DECLARATIONS

Funding

No external funding information was provided in the supplied project materials. If applicable, the authors should insert the funding agency and grant number.

Conflict of Interest

The authors should declare any financial or non-financial conflicts. If none exist, state: “The authors declare no conflict of interest.”

Ethics Statement

The current project uses secondary data sources as described in the supplied materials. Any use of personally identifiable social-media or incident data must follow applicable platform, institutional, and privacy requirements.

Data Availability

The project materials identify the Kaggle Road Accident dataset and a traffic-accident social-media dataset search. The final manuscript should provide the exact dataset version, access path, preprocessing code, and any licensing restrictions used in the completed study.

Author Contributions

All listed authors contributed to the project concept, methodology, implementation planning, evaluation design, and manuscript preparation. Specific contribution statements should be updated if the team's actual roles differ.

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Academic integrity note. All quantitative findings, dataset statistics, and model-performance claims must be populated from the team's actual experiments. The manuscript intentionally does not manufacture accuracy, F1, AUC, confusion-matrix counts, or statistical significance values that were not present in the supplied project materials.